RAYNARD TOM MAGEHEMA

Civil and Structural Engineer

raynard.magehema.24@ucl.ac.uk | +447417499150 | LinkedIn
London, UK | Registered Engineer (Tanzania, ERB No. 8359) | IStructE Student Member

Professional Profile

Civil engineer specializing in flood risk assessment, infrastructure resilience, and bridge engineering. Currently pursuing a PhD at University College London focusing on satellite-based flood vulnerability assessment of river-crossing infrastructure in data-scarce regions. Combines academic research with professional design experience in bridges, drainage, and hydrology.

Education

PhD Civil and Environmental Engineering

University College London, London, UK *2025 – Present*

- Flood fragility and vulnerability using remote sensing
- Chadwick Scholarship recipient (CEGE Department)
- Best PhD Impact Pitch; 3 Minute Thesis Winner (CEGE)

MSc Structural Engineering

University of Manchester, UK *2022 – 2023*

- Distinction; Equity and Merit Scholarship (146 applicants)

BSc Civil Engineering

Ardhi University, Tanzania *2015 – 2019*

- First Class Honours

Research Experience and Engagements

- Undergraduate Research – Adaptation of Prestressed Hollow Core Slabs in Tanzania.
- Research Collaborations:
  - DOI:10.1016/j.dim.2023.100044
  - Elsevier Article
  - DOI:10.5296/jss.v7i1.20488
- Master's Research Project – Blast Loading Fragility Assessment on Shallow Tunnels, University of Manchester.

Professional Experience

Assistant Lecturer in Civil Engineering

Ardhi University, Tanzania *2021 – Present*

- Teach Structural Analysis, Bridge Design, RC Design, Pavement Design, Hydrology
- Supervise 120+ undergraduate students annually
- Mentor final-year research and design projects

Team Leader – Bridge & Drainage Engineering

Selca Consult T Ltd *Oct 2024 – Dec 2024* *Road upgrade (1.58 km), Dar es Salaam, Tanzania*

- Led multidisciplinary design team for drainage and hydrology
- Prepared BOQs, material investigations, and reports
- Conducted flood risk assessments for infrastructure

Structural and Drainage Engineer

Selca Consult T Ltd / SABA Engineering PLC *Nov 2023 – Jan 2025* *TACTIC Project, Kigoma Region (World Bank funded)*

- Designed 35 m composite steel bridge (Eurocodes compliance)
- Performed flood risk and hydrologic analysis (multiple return periods)
- Designed drainage systems along 50+ km road alignment
- Prepared technical reports and supervision documents

Core Competencies

Engineering Design	**Software**	**Analysis**
Bridge structural design	MIDAS Civil	Structural design of bridges
Drainage systems	AutoCAD Civil 3D/ ArcGIS	Remote sensing and Hydrologic modeling
Road design	AutoCAD Civil 3D	Road Geometric Design
Reinforced concrete and steel structures design	Prota Structures/Midas GTS NX/Python	Structural design of buildings

Awards & Recognition

- 3 Minute Thesis Winner – UCL CEGE (2026)
- Best PhD Impact Pitch – CEGE Connect Festival (2025)
- Chadwick Scholarship – UCL, 2025
- Evaluation Partner - ABHOR Trellis Program UCL (2025)
- Equity & Merit Scholarship – University of Manchester
- Registered Professional Engineer – ERB Tanzania (No. 8359)
- IStructE Student Member

Languages & Strengths

English (Fluent), Swahili (Native)
Achiever, Communication, WOO, Harmony, Focus

References

Available upon request.